DIPLOMA PROGRAMME PRACTICE · HL · SET B

Biology / Paper 2

Worked mark scheme · 80 marks · 150 minutes

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each. Answer all of Section A and two of three Section B questions. Each extended response carries 15 content marks and one mark for overall quality.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 1 [12]

Testing two guide RNAs

(a) Calculate the intended-edit percentage and off-target percentage for each guide. [3]

1 A intended=156/200×100=78% and B intended=62%.

1 A off-target=9% and B off-target=2%.

1 A gives the greater intended percentage but B the lower measured off-target percentage.

(b) For guide A, calculate the possible range of colonies with both the intended edit and no detected off-target edit. Explain why one exact value cannot be found. [3]

1 At most all 156 intended edits are in colonies without the off-target edit.

1 At least 156−18=138 intended-edited colonies lack that off-target edit.

1 The overlap between the two positive groups was not recorded, so 138–156 are possible.

(c) Explain the roles of complementarity and Cas9 in the targeting mechanism, and why an imperfect off-target match could be relevant. [3]

1 The guide RNA base-pairs with a complementary target DNA sequence.

1 Cas9 recognizes an appropriate adjacent motif and cuts the target DNA.

1 A sufficiently similar second site can also be bound and cut, giving undesired sequence changes after repair.

(d) Assess the claim that B is the safer guide for all purposes and propose a stronger test. [3]

1 B has a lower detected off-target frequency at the one assayed locus in the surviving colonies.

1 Other sites or effects on survival are not represented, so a universal safety claim is unsupported.

1 Sequence additional off-target sites or genomes and measure survival with independent treated cultures and matched controls.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 2 [12]

Following labelled carbon in an alga

(a) Quantify the change in the share of label in 3-carbon acids between 2 s and 40 s in two ways. [3]

1 The share decreases by 76−12=64 percentage points.

1 Its relative decrease is 64/76×100≈84%.

1 These are changes in shares of recovered label, not absolute quantities of compound.

(b) Explain why early label in 3-carbon acids is consistent with the first fixation step of the Calvin cycle. [3]

1 Rubisco catalyses addition of CO_2 to RuBP.

1 The unstable six-carbon intermediate splits.

1 This yields two three-carbon phosphoglycerate molecules, one initially carrying the newly fixed carbon.

(c) Explain how light-dependent products enable some of this carbon to reach carbohydrate while fixation continues. [3]

1 ATP supplies energy and reduced NADP supplies reducing power in conversion to triose phosphate.

1 Some triose phosphate can be used to synthesize carbohydrates.

1 The rest contributes to regeneration of RuBP using ATP, allowing continued fixation.

(d) Explain why the percentages do not prove that every labelled atom passed directly from the first listed fraction to the second, and suggest an improvement. [3]

1 The data combine many molecules and alternative routes may contribute.

1 Each time point is normalized separately, masking total recovered radioactivity.

1 Resolve individual intermediates at more closely spaced times and measure absolute label recovery; accept another tracking design.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 3 [12]

Pressure differences and blood flow

(a) Calculate the pressure difference at rest and exercise and find the predicted exercise-to-rest flow ratio. [3]

1 Differences are 12.0 and 13.5 kPa.

1 Ratio=(13.5/0.75)/(12.0/1.00).

1 Predicted flow ratio is 1.50.

(b) Calculate predicted exercise output and stroke volume if exercise heart rate is 125 min^{-1}. [3]

1 Exercise output=5.0×1.50=7.5 $dm^3\ min^{-1}$.

1 Stroke volume=7.5/125=0.060 $dm^3\ beat^{-1}$.

1 This is 60 $cm^3\ beat^{-1}$ with the correct volume conversion.

(c) Explain how arterioles in active muscle can contribute to a lower vascular resistance and altered distribution of flow. [3]

1 Smooth muscle relaxation increases the arteriolar lumen diameter.

1 At a given pressure difference the wider lumen reduces resistance and increases local flow.

1 If vessels in other beds dilate less or constrict, a larger fraction of cardiac output reaches active muscle.

(d) Why would mean pressures be inadequate to determine the instant the aortic valve opens? State the necessary comparison and the consequence of valve closure. [3]

1 A mean hides within-cycle changes in pressure.

1 Opening occurs when left ventricular pressure exceeds aortic pressure, requiring simultaneous time-resolved measurements.

1 Closure prevents backflow into the ventricle when aortic pressure exceeds ventricular pressure.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 4 [12]

Following viral replication

(a) Estimate the burst size from the final increase above residual free particles. State the necessary assumption about infected bacteria. [3]

1 Increase=12010000−10000=12000000 particles cm^{-3}.

1 Estimated burst=12000000/(2.0×10^5)=60 particles per infected bacterium.

1 Assumes each counted infected cell completes the cycle and releases progeny once.

(b) Explain why a low early free-particle count does not mean viral nucleic acid is absent. [3]

1 Viral genomes can be inside infected host cells.

1 The assay samples free infective particles after removal of cells.

1 Assembly and release can lag behind intracellular genome replication.

(c) Distinguish lytic replication from a lysogenic association using the fate of the host and viral genome. [3]

1 In a lytic cycle viral genome and proteins are produced and progeny assembled.

1 Lysis releases particles and kills the host cell.

1 In lysogeny a prophage is maintained and copied with the host genome without immediate lysis.

(d) Suggest how to test whether a resistant surviving bacterial colony carries a prophage and evaluate a positive DNA result. [3]

1 Test colony DNA for a phage–host integration junction using suitable primers or sequencing.

1 Use known lysogenic positive and phage-free negative controls.

1 A junction supports integration but does not alone establish production of viable phage; induce and assay particles to test that.

SECTION B · EXTENDED RESPONSE
PAPER 2 · QUESTION 5 [16]

From enzyme shape to metabolic control

(a) Outline the role of activation energy and how an enzyme changes reaction rate. [3]

1 Reactants must reach a transition state requiring activation energy.

1 The enzyme provides an alternative pathway with a lower activation barrier.

1 At a given temperature more collisions can result in reaction per unit time.

(b) Explain how an enzyme–substrate interaction can generate specificity and catalysis. [5]

1 The active site has a particular three-dimensional arrangement of side chains.

1 Shape and chemical properties favour binding of suitable substrates.

1 Binding can induce a conformational change that improves interactions.

1 The enzyme can orient substrates or stabilize the transition state, facilitating bond changes.

1 Products leave and the enzyme can catalyse another cycle.

(c) Explain feedback inhibition and how a student could distinguish reversible competitive inhibition from a reduction in maximum catalytic capacity. [7]

1 A downstream product binds a regulatory site on an earlier enzyme.

1 A conformational change reduces that enzyme's activity.

1 As product accumulates, flux through the pathway falls, conserving substrates or energy.

1 When product level falls, inhibition is relieved and flux can rise again.

1 Measure initial rates across a range of substrate concentrations with and without inhibitor while controlling enzyme concentration and temperature.

1 Under simple reversible competitive inhibition, sufficiently high substrate can approach the same maximum rate.

1 A persistent lower plateau at saturating substrate supports reduced catalytic capacity; the pattern alone need not identify the exact binding site.

(Quality) Quality of the complete extended response [1]

1 Award one mark when the complete response communicates the relevant biological reasoning clearly and concisely, with logical connections and little repetition. Assess this across the parts, independently of the content score.

SECTION B · EXTENDED RESPONSE
PAPER 2 · QUESTION 6 [16]

Water movement from soil to air

(a) Distinguish solute potential from pressure potential and relate them to total water potential in a cell. [3]

1 Dissolved solutes lower water potential relative to pure water, giving a negative solute potential.

1 Pressure exerted on cell contents by the wall usually provides a positive pressure potential in a turgid cell.

1 Ignoring gravity and other components, total water potential is the sum of solute and pressure potentials.

(b) Explain water uptake from soil into a root and onward delivery to xylem when the gradient is favourable. [5]

1 Root hairs increase contact area with soil solution.

1 Solute accumulation can lower root-cell water potential.

1 Water moves across a partially permeable membrane by osmosis down the water-potential gradient.

1 Water can move across cortex through walls or through cell contents.

1 Selective passage through endodermal cells regulates entry of solutes into the vascular tissue; accept an accurate explanation of the Casparian strip.

(c) Explain the cohesion–tension mechanism and how a prolonged drought can limit carbon gain. [7]

1 Water evaporates from moist internal leaf surfaces.

1 Water vapour diffuses out through stomata down its gradient.

1 Loss of water generates tension in the leaf xylem.

1 Cohesion maintains continuity of water columns as tension is transmitted down the xylem.

1 Water is pulled upward to replace water lost at leaves.

1 Under water deficit, stomata can close, reducing transpiration.

1 Closure restricts CO_2 entry and can lower photosynthetic carbon fixation.

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SECTION B · EXTENDED RESPONSE
PAPER 2 · QUESTION 7 [16]

Chromosomes, linkage and variation

(a) Outline how homologous chromosomes differ from sister chromatids. [3]

1 Homologues carry corresponding loci and one member comes from each parent.

1 They may have different alleles at a locus.

1 Sister chromatids arise by replication of one chromosome and initially carry matching DNA sequences, apart from copying errors.

(b) Explain the events in meiosis that allow gametes to receive one allele at each locus and different combinations at linked loci. [5]

1 DNA replicates once before meiosis begins.

1 Homologous chromosomes pair in prophase I.

1 Crossing over between non-sister chromatids can exchange segments between the loci.

1 Homologues separate in the first meiotic division, reducing chromosome number.

1 Sister chromatids separate in meiosis II, generating haploid products.

(c) Discuss how test-cross progeny can be used to infer linkage, with limits on the inference. [7]

1 Cross the double heterozygote with an individual recessive at both loci.

1 Provided phenotypes reliably reveal genotype, each offspring identifies the heterozygote's gamete type.

1 Independent assortment with equal transmission and survival predicts four equally frequent classes.

1 An excess of the two parental classes supports linkage.

1 Recombinant percentage equals the combined nonparental counts divided by total progeny times 100.

1 Double crossovers can restore a parental marker arrangement and cause underestimation of physical separation.

1 Differential viability or small samples can distort observed frequencies; use sufficient progeny and test goodness of fit or score earlier stages.

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